

Inclusive AI Fashion Stylist with Measurement and Color-Blind Safety Analysis

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Project Proposal Report

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DECLARATION

DECLARATION

I declare that this is my own work, and this proposal does not knowingly incorporate any material previously submitted for a degree or diploma at any other university or higher education institution, nor does it contain any material that has been previously published or written by another person apart from where proper acknowledgment is given in the text.

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The above candidate is conducting research for their undergraduate Dissertation under my supervision.

Name	Date	Signature
Narmada Gamage	15/03/2026	

ABSTRACT

Online fashion shopping has grown considerably, providing consumers with access to numerous clothing items. Yet it is still difficult to choose clothing items that match personal needs, size, and occasion. The majority of current fashion recommendation systems are based on collaborative filters and visual similarities, which are unable to fully understand customers' natural language needs and requirements simultaneously. As such, customers are often recommended inappropriate items, leading to dissatisfaction and increased rates of returned products in online shopping platforms. The current research aims to develop a text-driven occasion-aware fashion recommendation system that can enhance the level of personalization and accuracy in fashion recommendations for online fashion shoppers in Sri Lanka. The system will enable customers to input their clothing needs in the form of natural language text, specifying their preferred style, colors, and occasion, whether it is for a wedding, party, or office event. A Natural Language Processing (NLP) module is used to process the user's input based on text embedding, which enables the identification of relevant attributes of clothing items from the dataset based on semantic similarity. Additionally, the proposed fashion recommendation system includes a size filtering component that ensures the recommended clothing items are available in the user's predicted size. Furthermore, the proposed fashion recommendation system includes a context-aware component that ranks clothing items based on the context provided by the user. Overall, the proposed fashion recommendation system is able to incorporate the benefits of NLP-based preference analysis, context-awareness, and size prediction to produce accurate fashion recommendations. The proposed fashion recommendation system is likely to produce an intelligent fashion recommendation prototype that is able to improve user satisfaction, reduce return rates, and incorporate culturally relevant fashion recommendations in the growing e-commerce market of Sri Lanka. Keywords: Fashion Recommendation System, Natural Language Processing, Personalization, Size Prediction, E-commerce.

ACKNOWLEDGEMENT

Such a research effort does not come by itself. It results from guidance provided, knowledge imparted, efforts put in, and faith bestowed most often by those who ask for nothing in return. I would like to acknowledge those individuals whose contributions have helped make this research effort possible.

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I would specifically like to express my appreciation to my research partner, whose tireless efforts in developing the body measurement and size prediction module were essential to the integrated effort. His professionalism, reliability, and commitment to our joint effort are appreciated. This research would not have been possible without the over three hundred individuals from Sri Lanka who volunteered their time to participate in our survey. They provided the honest and genuine input that turned this system from a technical exercise into a solution that is relevant to real-world problems.

To my family, who have been with me throughout every doubt and every deadline, your sacrifice has not gone unnoticed, and this achievement is as much yours as it is mine.

TABLE OF CONTENT

DECLARATION	2
ABSTRACT	3
ACKNOWLEDGEMENT	4
TABLE OF CONTENT	5
LIST OF TABLES	6
LIST OF FIGURES	7
LIST OF ABBREVIATIONS	8
1. INTRODUCTION	9
1.1 Background and Context.....	9
1.2 Literature Review Discuss existing products/services	11
1.3 Research Gap	13
2. OBJECTIVE	15
2.1 Main Objective.....	15
2.2 Specific Objectives	15
3. METHODOLOGY	16
4. HIGH-LEVEL SYSTEM ARCHITECTURE	17
5. USER REQUIREMENTS.....	17
6. COMMERCIALIZATION PLAN	20
7. BUDGET AND JUSTIFICATION.....	23
8. WORK BREAKDOWN STRUCTURE (WBS).....	24
9. GANTT CHART.....	25
10. REFERENCES LIST	26
11. APPENDICES	27

LIST OF TABLES

Table 1 Abbreviations table	8
Table 2 Gap analysis table	14
Table 3 Stakeholder table.....	18
Table 4 Budget Summary.....	23

LIST OF FIGURES

Figure 4.1high-level system architecture diagram.....	17
Figure 6.2 Commercial plan	22
Figure 8. 3 WBS	24
Figure 9.4 Gant Chart	25
Figure 11.5 Occasions wise clothes	27
Figure11. 6 Color wise chart.....	27
Figure 11.7 AI system chart	27
Figure11. 8 Detail diagram	28

LIST OF ABBREVIATIONS

Table 1 Abbreviations table

Abbreviation	Meaning
AI	Artificial Intelligence
API	Application Programming Interface
B2C	Business to Consumer
CSV	Comma-Separated Values
DSR	Design Science Research
FRS	Fashion Recommendation System
ML	Machine Learning
NLP	Natural Language Processing
UI	User interface
UX	User Experience

1. INTRODUCTION

1.1 Background and Context

Choosing fashion has turned into a more complicated challenge for today's shoppers. Individuals need to take into account various elements like body shape, personal taste, color choice, and the event when selecting apparel. As online fashion retail expands quickly, shoppers face thousands of clothing choices, complicating their decision-making process. Consequently, numerous users have difficulty locating clothing that suits them properly and aligns with their tastes. This issue impacts individual shoppers and online sellers alike, as bad clothing selections frequently result in disappointment and returns. Research on fashion recommendation systems indicates that existing platforms continue to struggle with effectively grasping user preferences and delivering tailored suggestions, particularly in complex fashion scenarios and styling combinations [3], [4], [5].

The global e-commerce apparel market is projected to rise to 1.62 trillion USD by 2035 from 808.1 billion USD in 2026. The fashion returns online are 25-40%, which is much more compared to in-store returns, which range from 8-10%, out of which 60-70% returns are due to size/style mismatch [1]. Simultaneously, the e-commerce market in Sri Lanka is quickly growing, with projections suggesting that the B2C e-commerce segment may hit around \$3.2 billion by 2029, underscoring the increasing significance of smart digital shopping solutions [2]. With the ongoing expansion of online fashion shopping both globally and locally, enhancing the precision of recommendations and personalization is becoming ever more crucial.

Current fashion recommendation systems usually depend on collaborative filtering, content-based filtering, or a combination of both methods. These systems examine previous buying habits, reviews, or product attributes to suggest clothing pieces [3], [4]. Recent studies have investigated AI-based and deep learning systems, multimodal recommendation methods, and context-sensitive models that integrate image, text, and user interaction information [5], [6]. Certain contemporary solutions employ natural language processing (NLP) to comprehend fashion descriptions or trends, allowing users to find clothing through descriptive text queries [7], while others utilize vision-language models and multimodal AI to develop smart fashion assistants that can interpret both images and text [11].

In spite of these improvements, numerous constraints continue to persist in existing systems. Numerous platforms concentrate solely on suggesting single clothing pieces instead of entire outfits, restricting the practicality of the recommendations [6]. Certain systems depend significantly on past data and user engagement, posing challenges for new users who have minimal data (the cold-start problem) [3]. Moreover, many current solutions continue to be insufficiently integrated, failing to merge user preferences, textual descriptions, body dimensions, and contextual factors like occasion or style needs [4], [5]. While studies have examined size prediction models and fit suggestion methods utilizing machine learning and deep learning, these models are frequently applied independently from recommendation systems, diminishing their overall impact in practical scenarios [10]

To tackle these issues, this study introduces a smart text-based fashion recommendation system that combines various elements to deliver more tailored clothing recommendations. The suggested system enables users to articulate their clothing requirements in natural language, including details about the event, desired colors, and style choices. An NLP module evaluates the text input and derives organized attributes to assist the recommendation procedure. Moreover, the system includes a body measurement and size forecasting module to enhance the precision of size suggestions, assisting users in choosing garments that fit well. The system strives to provide more pertinent and useful fashion suggestions by integrating user preference analysis, contextual comprehension, and size forecasting.

This study also corresponds with wider objectives of sustainable development and innovation. Minimizing erroneous purchases and returns of products can aid in lowering logistics waste and environmental effects, contributing to Sustainable Development Goal (SDG) 12: Responsible Consumption and Production. Moreover, the study supports SDG 9: Industry, Innovation, and Infrastructure by advancing the creation of smart digital technologies in the fashion e-commerce industry. From a research standpoint, the study is categorized under the Artificial Intelligence, Data Science, and Intelligent Systems research cluster, as it combines machine learning, natural language processing, and recommender system technologies to address a practical issue in online retail

1.2 Literature Review Discuss existing products/services

This literature review examines recent research (2021–2025) regarding fashion recommendation systems and their methodologies. It emphasizes significant constraints like restricted natural language input management, absence of situational awareness, and inadequate size integration, which drive the suggested text-based, context-aware recommendation system for Sri Lanka [3]–[6]

Existing Research Approaches

Fashion recommendation systems have transitioned from conventional filtering approaches to sophisticated AI-powered techniques. Chakraborty et al. offer an extensive overview of collaborative filtering, content-based and hybrid methods, along with image parsing, attribute identification, and outfit compatibility modeling in FRS. They emphasize that although visual elements prevail in early systems, there is dependence on resemblance instead of profound compatibility or user context [3]. Deldjoo et al. propose a classification that includes outfit generation, pair recommendation, compatibility prediction, and personalization, highlighting that multimodal approaches that integrate visual attributes are successful, yet the management of varied contexts is still constrained [4]. Shirkhani et al. highlight that fashion compatibility involves greater complexity than similarities in other areas, criticizing various systems for overlooking contextual elements like occasion, and personal preferences [5]. Prompt-based retrieval techniques employ CLIP-like models alongside text prompts (such as color, style, occasion) to address occasions and needs directly within customized workflows [8]. GAN-based context-aware systems like CRGAN focus on particular demographics (e.g., young females in Sri Lanka) by incorporating occasion labels; however, they frequently fail to achieve a comprehensive fusion of text and size [9]. Foundational models predicting size and fit based on purchase history and body measurements tackle size issues, yet they consider size independently from style or context [10].

Existing Products / Services

E-commerce sites like Amazon, Daraz, Myntra, and Zalando utilize a mix of collaborative filtering and visual similarity to generate recommendations, which include size guides and “customers also purchased” prompts. Nonetheless, these platforms do not possess comprehensive natural language parsing to grasp contexts and subtle user needs [4], [5]. Tools powered by AI such as Fit Analytics and Fit Finder enhance return rates (10–20%) by

forecasting sizes yet fail to include details about style or occasion [10]. Personalized size and style advisors like True Fit and Style DNA provide user profiles but are still restricted in their integrated text-based occasion matching capabilities [10]. Graph-driven trend recommendation systems (similar to Pinterest and inspired by PinSage) offer visual and trend-oriented suggestions yet fail to integrate user text with context or scale for accurate personalization [6], [11]

Limitations of Existing Systems

In spite of these developments, significant constraints remain. Many systems primarily depend on visual attributes while only marginally integrating natural language inputs for individual preferences, restricting the extent of personalization and contextual subtleties [3], [5]. Even models that are “context-aware” typically regard context as fixed labels instead of actively integrating user-generated text with detailed context like event details [6], [9]. Preferences for size and style are typically addressed independently, leading to discrepancies and increasing return rates [8]. Moreover, training datasets focused on the West overlook South Asian preferences, such as climate, cultural events, and body shapes, restricting local significance [6], [9]. Subjectivity, cold-start problems, and insufficient explainability continue to pose significant challenges, as recent NLP and prompt-centric methods infrequently consider scale and local cultural influences [7], [8], [11].

These constraints highlight a distinct research gap: the necessity for a completely integrated, text-oriented fashion recommendation system that unifies NLP-driven preference interpretation, awareness of occasion/context, size customization, and adaptation to local culture while delivering explainable, user-focused results. Bridging this gap would enhance personalization, lower returns, and boost user satisfaction in growing e-commerce markets like Sri Lanka.

1.3 Research Gap

Choosing fashion can frequently be difficult for shoppers around the globe. Selecting appropriate attire according to the event, body shape, dimensions, and individual taste can be complicated and take a lot of time. Digital platforms provide numerous choices, which can confuse users. While current fashion recommendation systems (FRS) assist users, they frequently do not fully incorporate text-based preferences, awareness of occasion/context, and size personalization, particularly in culturally unique markets such as Sri Lanka [3], [4]

The existing FRS mainly overlook these aspects:

- Advanced NLP for flexible text inputs – Users express requirements in everyday language, for instance, “comfortable cotton saree for a humid Colombo wedding, modest but stylish”. Many systems depend on organized tags, visuals, or previous purchases rather than interpreting subtle user text [4].
- Comprehensive merging of occasion/context with text and dimensions – While some models identify occasions (such as weddings or office events) or weather, very few effectively integrate these with user text and size as rigid constraints [5].
- Equitable, gender-inclusive personalization – Male users are lacking; the majority of systems emphasize women or visual compatibility [4], [5]

Existing systems fall short due to structural limitations:

- Dominance of image/visual and collaborative filtering – CNN-based or CF-based models focus on visual compatibility or past behaviors [3], [4], ignoring subjective text inputs and dynamic context [8].
- Limited context integration – Even context-aware models treat occasion/weather as static tags, not dynamically fused with NLP-analyzed user text [5]. Complex queries like “comfortable yet elegant for party” are poorly handled.
- Separated handling of size/fit – Size prediction is often standalone or presented as basic charts, not integrated with style and preference matching [3].
- Data and domain biases – Western datasets dominate, overlooking cultural nuances. Emerging prompt-based approaches rarely integrate local sizes or surveys [4].
- Subjectivity and dynamics – Fashion is personal and trend-sensitive. Static retrieval pipelines fail on unseen attributes or rapid shifts [6].

The proposed system is novel because it:

The proposed system improves fashion recommendations by incorporating NLP embeddings, occasion boosting, and size filtering for more personalized and understandable fashion recommendations. For instance, the system can provide fashion recommendations for items [4], [6]. Moreover, the system includes local Sri Lankan survey results to consider user preferences for age, gender, occasions, and fabrics, thereby improving tagging in the datasets and avoiding cultural bias [5], [8]. Furthermore, the system includes fashion recommendations for both ladies and gents, addressing the issue of gender bias in fashion recommendations, where male users are often ignored. Overall, the system, based on NLP embeddings, context awareness, and size integration, targets the major causes of fashion returns in online shopping, such as improper fits and styles, for improved user satisfaction and sustainability .

Table 2 Gap analysis table

Gap Area	Current Approaches	Limitations / Insufficiency	Evidence / References
Multi-Modal Integration	Multimodal FRS combining image features and metadata	Visual, textual, and size information are often handled separately, limiting full personalization	Deldjoo et al., 2023 [4]; Liu et al., 2025 [11]
Occasion/Context	Occasion-aware models, context tagging systems	Context often treated as static labels (e.g., wedding, office) rather than dynamically integrated with user preferences	Shirkhani et al., 2023 [5]; Pathirana et al., 2025 [9]
Free-Text / Conversational Input	NLP-based fashion retrieval and prompt-based recommendation	Difficulty interpreting complex natural language queries combining style, occasion, and constraints	Mallik et al., 2025 [7]; CEUR-WS, 2024–2025 [8]
Size & Fit Accuracy	Size prediction models and size recommendation tools	Size prediction often separate from style and context matching, causing recommendation mismatch	Sheikh et al., 2019 [10]
Gender Balance	Fashion recommendation models mainly trained on women's fashion datasets	Male users are often underrepresented, leading to limited recommendation diversity for men	Deldjoo et al., 2023 [4]; Shirkhani et al., 2023 [5]

2. OBJECTIVE

2.1 Main Objective

To develop and evaluate a text-driven, occasion-aware clothing recommendation system that integrates natural language user inputs (preferences, requirements, and occasion descriptions), personalized size filtering, and an online fashion dataset to provide accurate, explainable, and culturally relevant dress recommendations for both ladies and gents in the Sri Lankan e-commerce context.

2.2 Specific Objectives

These 5 specific objectives are aligned with the research problem (lack of integrated text + occasion + size personalization in fashion RS) and map to the individual components of your group project:

1. Collect and analyze local user data through a survey of at least 500 respondents in Sri Lanka to capture demographics (age, gender), preferences (colors, styles), occasion-specific needs (e.g., weddings, office, festivals), and requirements (e.g., breathable for humidity, budget constraints), providing a culturally adapted foundation for dataset tagging and system validation.
2. Design and implement a natural language processing (NLP) module using text embeddings (e.g., sentence-transformers) to analyze user-input text descriptions of preferences, occasions, and requirements, computing semantic similarity scores between user queries and dress descriptions in the dataset to enable nuanced matching beyond keyword-based approaches.
3. Integrate size recommendation from the teammate's function as a hard filter in the recommendation pipeline, ensuring all suggested dresses are available in the user's predicted/recommended size range, thereby reducing fit-related mismatches and improving overall relevance.
4. Develop an occasion-aware ranking mechanism that boosts scores for dresses whose descriptions or tags match the user-specified occasion (e.g., "wedding", "party", "office") while incorporating local contextual factors (e.g., tropical fabric preferences), to produce contextually appropriate top recommendations.
5. Evaluate the system's performance using both quantitative metrics and qualitative user feedback (e.g., relevance ratings 1–5 from 500+ survey participants), comparing it against baselines (e.g., random, simple keyword matching) to demonstrate at least 20–30% improvement in personalization effectiveness.

3. METHODOLOGY

Research Approach

This project uses Design Science Research (DSR) to develop a text-driven, occasion-aware fashion recommendation system. DSR is suitable as it addresses high return rates and poor personalization in Sri Lankan e-commerce, integrating user text, occasion/context, and size for actionable recommendations through iterative design and evaluation cycles.

Technical Methods

The system uses NLP (Sentence-Transformers) to convert user queries and product descriptions into embeddings, matched via cosine similarity. Size filtering ensures items fit using a teammate's prediction function, and an occasion-aware ranking prioritizes items matching user-specified events and local context. Evaluation combines Precision@5, Recall@5, NDCG@10, and qualitative feedback from 500+ users.

Tools and Platforms

Python handles ML/NLP, Pandas/Scikit-learn process data and calculating similarity, SQLite/PostgreSQL store datasets, and Google Colab enables prototyping. Flask/FastAPI serve recommendations, Git/GitHub ensures version control, and datasets come from Kaggle or internal CSVs.

Data Collection, Ethics, and Validation

Data is collected from 500+ Sri Lankan survey participants and product datasets. Ethical measures include anonymity, consent, and gender-balanced recommendations. Validation uses both quantitative metrics and qualitative user feedback to ensure accuracy and satisfaction.

4. HIGH-LEVEL SYSTEM ARCHITECTURE

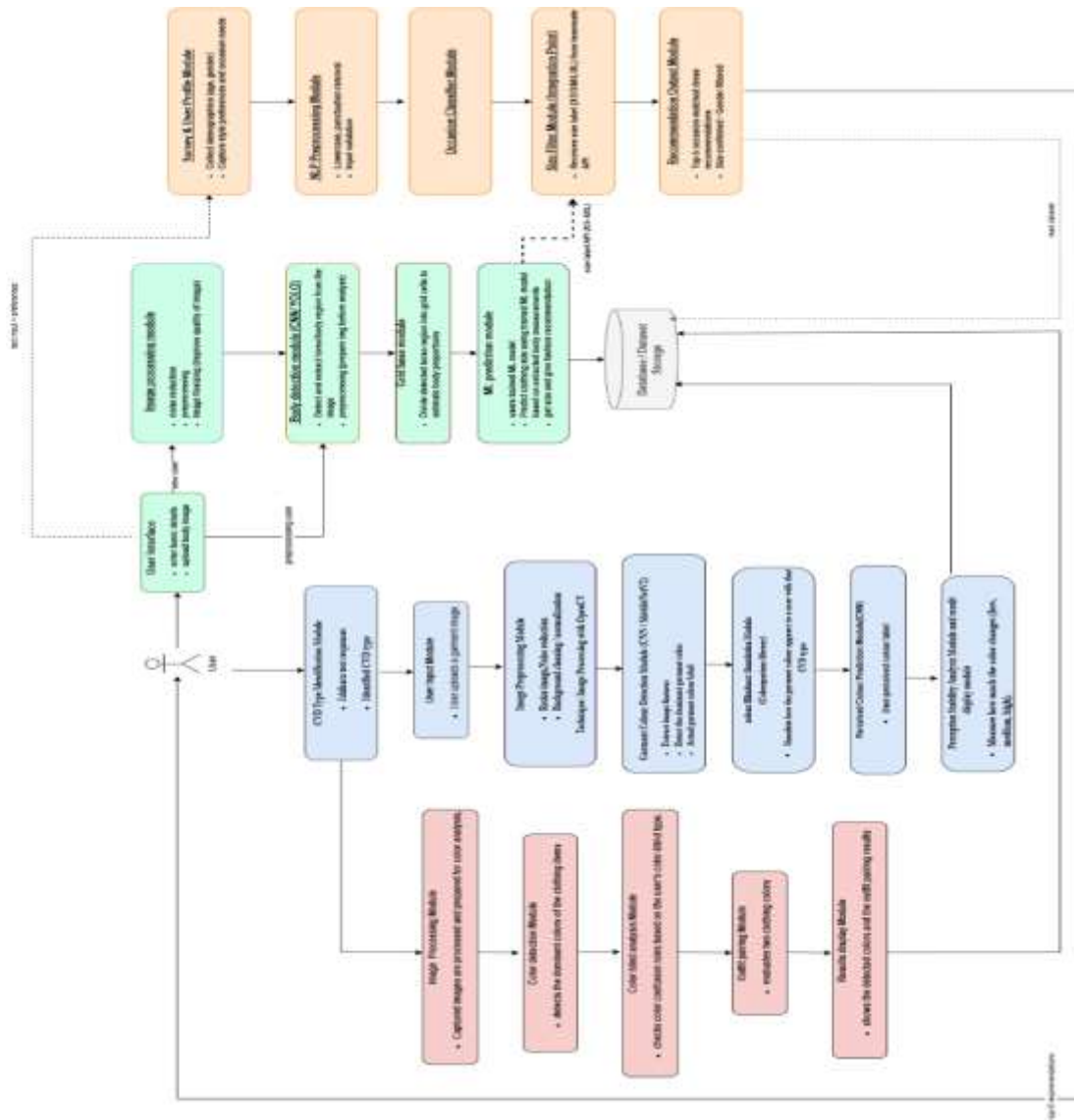


Figure 4.1 High level system architecture

User Input: Receives the user's query, size, and occasion **Module:** Converts the user query into a structured format using Sentence Transformers. **Semantic Matching Engine:** Retrieves relevant products from the dataset based on style, size, and occasion. **Size Filtering:** Ensures recommended products match the user's predicted size. **Occasion-Aware Ranking:** Ranks products according to the specified occasion. **Recommendation Output:** Displays the recommended products to the user. **Feedback Module:** Collects user ratings and survey data to improve and retrain the system. **Admin / Cloud Server:** Manages dataset updates, scalability, and system security.

5. USER REQUIREMENTS

Stakeholder identification

Table 3 Stakeholder table

Stakeholder	Description	Primary Interest	Interaction Type
End User — Ladies	Female consumers in Sri Lanka shopping for occasion-appropriate clothing online or via a prototype interface	Receive accurate, stylish, occasion-matched dress recommendations that reflect personal preferences and are culturally appropriate	Direct — primary users of the recommendation interface
End User — Gents	Male consumers in Sri Lanka seeking outfit guidance for professional, casual, cultural, or special occasion needs	Receive relevant menswear recommendations without needing fashion expertise, matching their stated occasion and preference text	Direct — primary users of the recommendation interface
Research Team	The student research group developing and evaluating the system, including the clothing recommendation member (this research) and the size recommendation member (teammate)	Validate system performance against defined metrics; fulfill academic research objectives; demonstrate prototype viability	Direct — system designers, developers, and evaluators
Survey Respondents	500+ Sri Lankan participants who complete the preference survey and provide system evaluation feedback	Contribute to culturally relevant system design; receive a system that accurately reflects local fashion needs and occasions	Direct — data contributors and evaluators in the user study

Functional requirements

FUN 1 -Accept Free-Text Inputs: Users can provide unstructured natural language inputs describing their clothing preferences, occasion, style.

FUN 2 -Based Semantic Processing: The system shall implement Sentence-Transformers or similar embedding models to process user text, understanding meaning beyond simple keywords.

FUN 3 - Size and Fit Personalization: The system shall integrate size prediction or filtering as a core requirement.

FUN 4 - Product Matching and Filtering: Recommendations must incorporate semantic matching with the product dataset

FUN 5 -Occasion-Aware Ranking: The system shall boost items that match the specified occasion (e.g., wedding, office, festival) and apply local contextual knowledge

FUN 6 -User Feedback Collection: The system must allow users to provide ratings and surveys about recommendation relevance, style, occasion match, and size accuracy. Feedback will be used to refine models iteratively.

FUN 7 -Gender Inclusivity: Recommendations must support both male and female users,

FUN 8 -Integration with Fashion Dataset: The system shall maintain up-to-date product information

Non-Functional Requirements

1. Security: User input and survey data must be stored securely. No personal identifiers will be retained, and data usage will comply with ethical standards.
2. Performance: Recommendations should be generated in under 2–3 seconds to provide a smooth user experience, even with large datasets.
3. Usability: The system interface must be intuitive and user-friendly, allowing users to easily input preferences, understand recommendations, and provide feedback.
4. Scalability: The architecture should support increasing numbers of users and dataset growth without significant performance degradation.
5. Reliability: The system must operate consistently and robustly, minimizing downtime or errors in delivery.
6. Explainability: Each recommendation should provide transparent reasoning so that users understand why a particular item was suggested, increasing trust and engagement.
7. Maintainability: The system should allow easy updating of models, datasets, and rules, ensuring that new trends or fashion items can be incorporated quickly.

6. COMMERCIALIZATION PLAN

Target Market and Customer Persona

The proposed fashion recommendation system is designed to target the online buyers of Sri Lanka, belonging to the age group of 18 to 45 years, and buy clothing for various personal, social, and professional needs. This system would target different people as customers. Young professionals, belonging to the age group of 22 to 35 years, belong to both genders. This group requires efficient and style-oriented fashion recommendations for various needs, including office, casual, and party wear. They would also appreciate the precision and cultural appropriateness of the clothing. Event and wedding buyers belong to the age group of 25 to 40 years and require clothing for specific events, such as sarees, dresses, and suits. This group would appreciate the precision of the recommendations but would also expect to pay for the same. Fashion enthusiasts belong to the age group of 18 to 30 years and would appreciate the recommendations that are both culturally appropriate and trendy.

Value Proposition

The system provides a unique combination of text-based input, personalization of size and fit, and occasion-based suggestions. The user can input their clothing needs in natural language, and the system offers hyper-personalized suggestions with an explanation of why a particular item is suggested. The cultural factor is also included in this system, which is specific to Sri Lanka.

Revenue Model

The revenue model of the system consists of multiple revenue streams. The subscription-based revenue stream enables premium users to access advanced personalization features through subscription fees. The platform integration fees are applicable to e-commerce platforms that integrate the system into their services. The affiliate or commission revenue stream enables the generation of revenue through sales of recommended products. The analytics revenue stream enables the generation of revenue through anonymized consumer insights and trend analysis provided to retailers.

Cost Estimation

The major cost factors include the development costs for the NLP module, the recommendation engine, and UI/UX design. Cloud infrastructure costs include hosting, storage, and computation for real-time recommendation. Data collection costs include conducting surveys and preparing the dataset. Maintenance and update costs include retraining the model and monitoring the system. Marketing costs include promotion and customer acquisition. The costs are estimated based on the features and scalability planned.

Pricing Strategy

The system operates on a tiered pricing model. There is a free tier for basic recommendations, but they are not personalized. There is a premium tier for NLP-driven, occasion-aware, and size-integrated recommendations, which requires a subscription. There is also an enterprise tier for e-commerce platforms, offering API, analytics, and support.

Competitive Advantage

The system stands out from the rest based on its integration with personalization, which includes text, occasion, and size. Additionally, the system has been customized to suit the needs of Sri Lankan festivals. The system allows for explainable recommendations, which increases the trust of the users. The system allows for gender diversity since it works for both men and women.

Intellectual Property Considerations

The intellectual properties include copyright and patent for the software and algorithm. The dataset licensing allows for the ethical use of the product metadata and survey. Trademark for the branding, which includes the name and logo, can be used for the system.

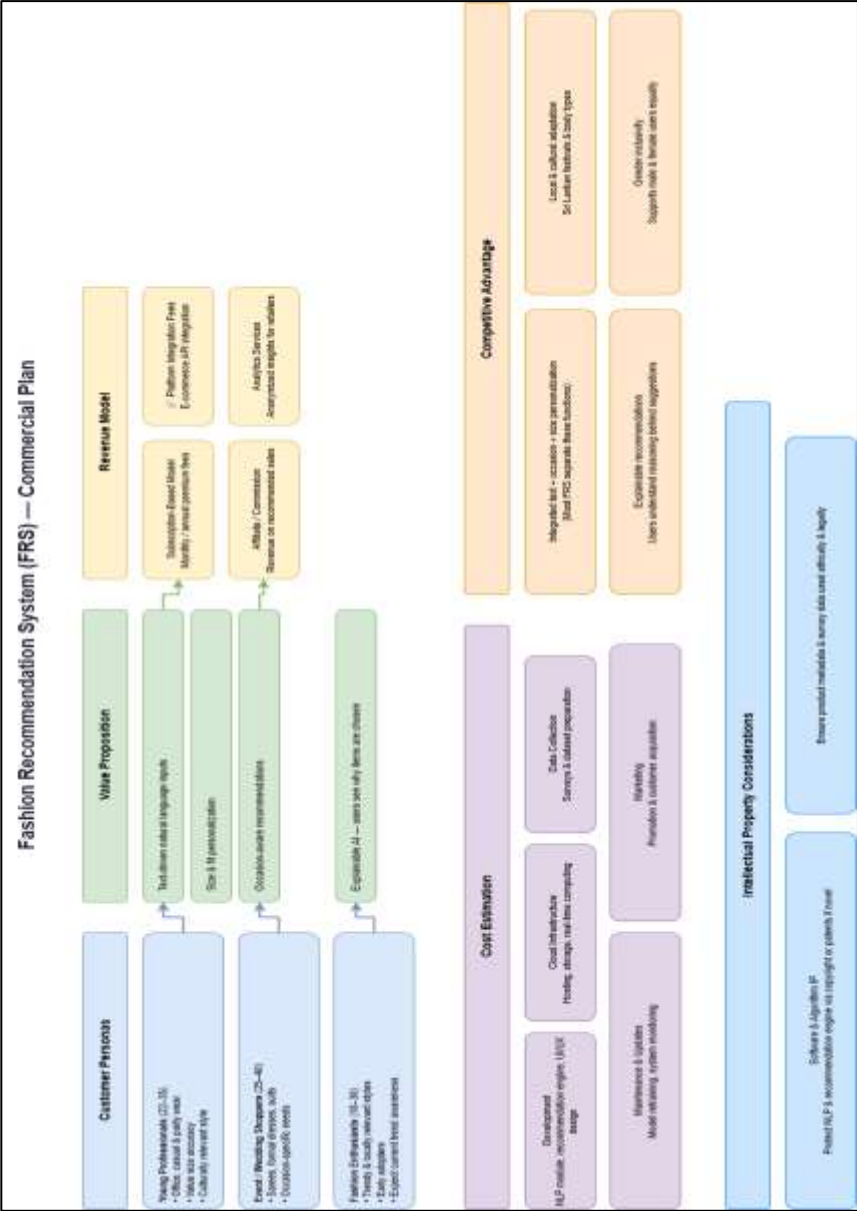


Figure 6.2 Commercial plan

7. BUDGET AND JUSTIFICATION

Table 4 Budget Summary

Technology / Resource	Total (LKR)
Hardware & Infrastructure	
Laptop / personal computer	0
Cloud & Compute Services	
Cloud storage and hosting(firebase AWS-10 months)	55,000.00
Wifi/ broadband connection (10 months)	25,500.00
Dataset & Data Resources	
Data collection materials, test & documentation	10,500.00
Contingency & miscellaneous	
Unexpected expenses	7,250.00
Total	97,250.00

8. WORK BREAKDOWN STRUCTURE (WBS)

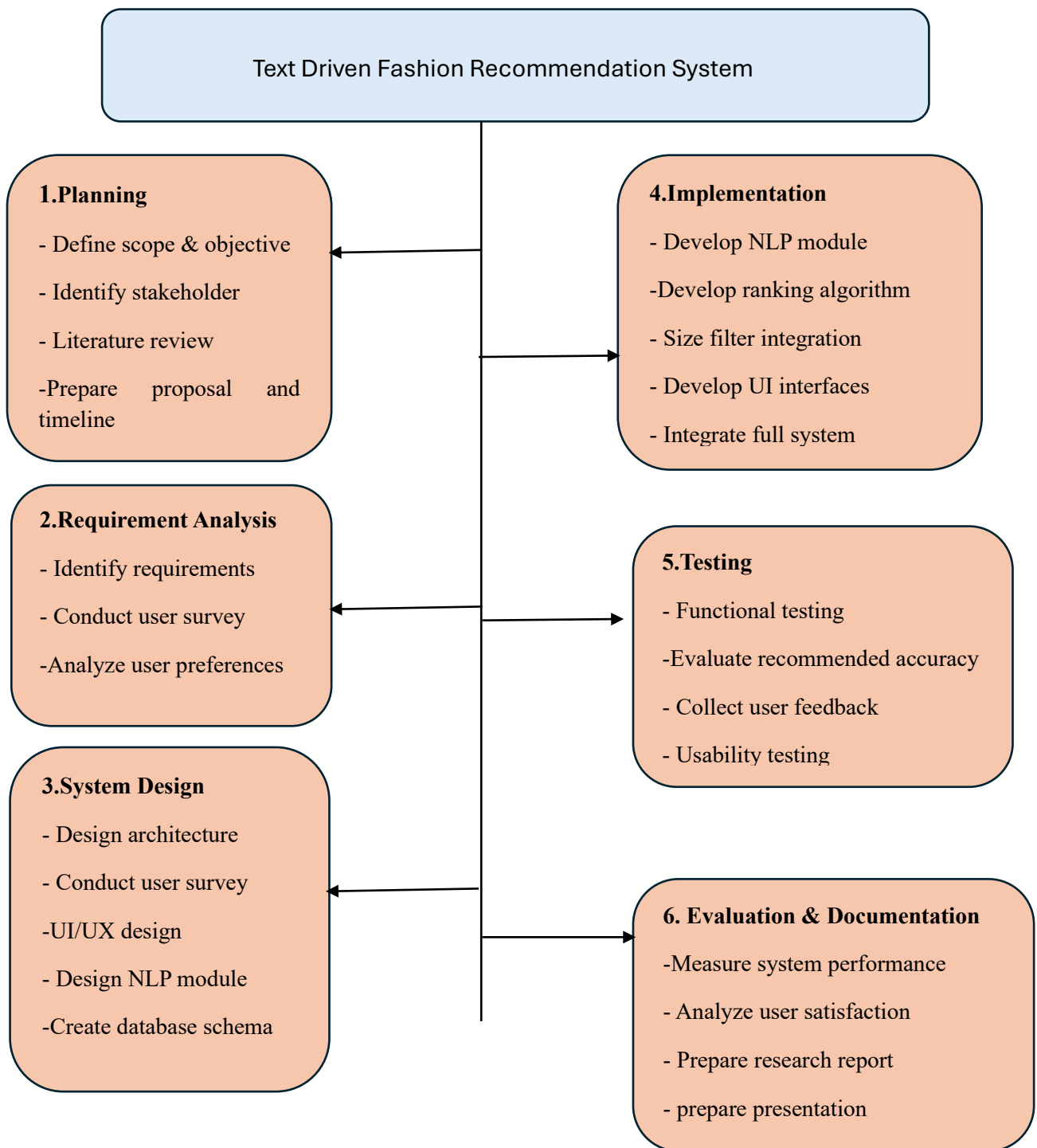


Figure 8. 3 WBS

9. GANTT CHART

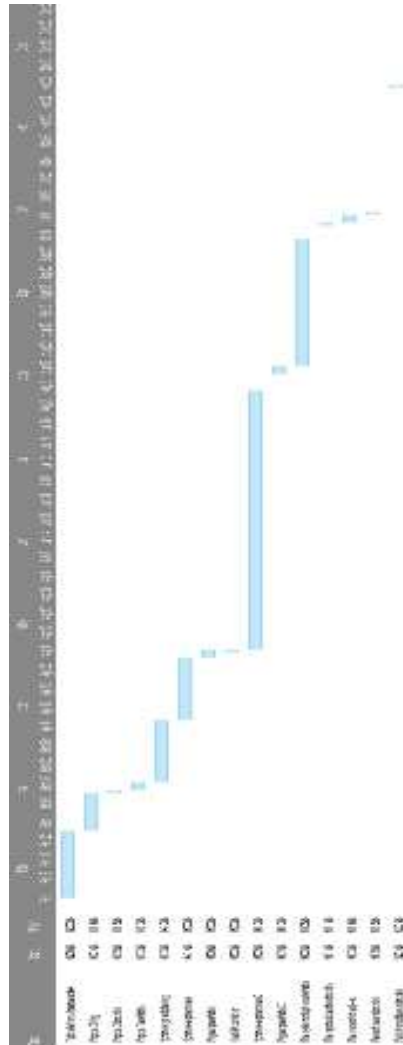


Figure 9.4 Gantt Chart

<https://drive.google.com/drive/u/0/folders/1ly9Y52QmGQ342dv4HI6odvbygFQqMOFR>

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11.APPENDICES

- Survey instruments

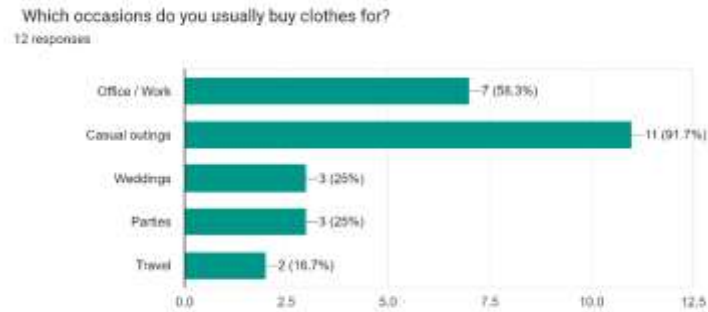


Figure 11.5 Occasions wise clothes

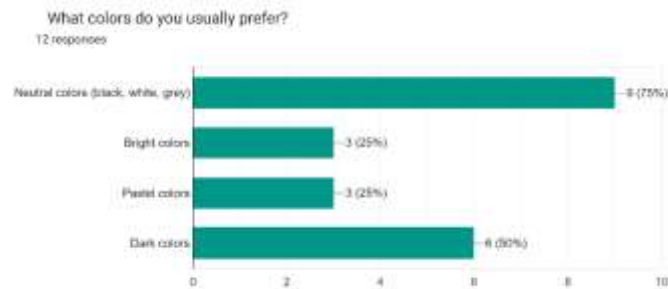


Figure11. 6 Color wise chart

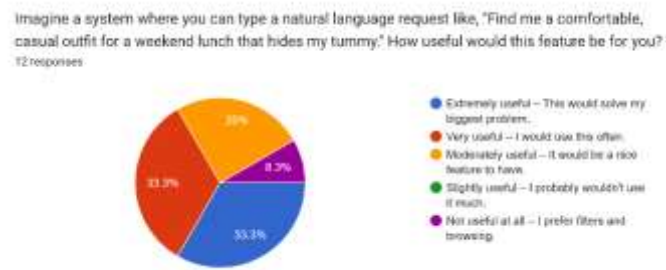


Figure 11.7 AI system chart

https://docs.google.com/spreadsheets/d/1xBXEGr9Zj6uikABfyLxROg_h2DXUVXPwi8G_E-3V2Ao/edit?resourcekey&usp=forms_web_b&urp=linked#gid=773680301

https://docs.google.com/forms/d/1cuDhg8F809h1T70Q9plBniDG0gb3dWiT-ecxo_RT7M/edit#responses

- Detailed Diagram

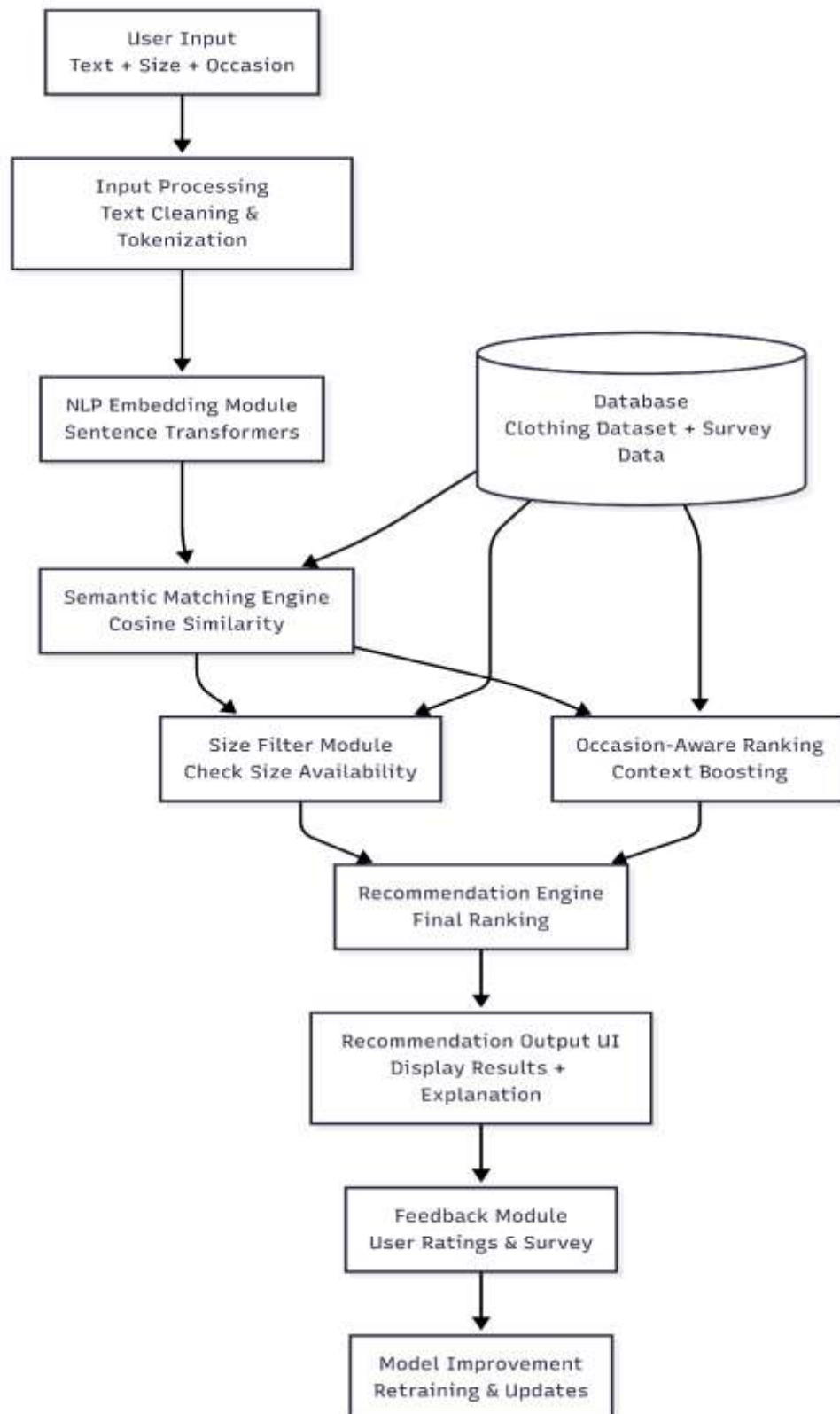


Figure11. 8 Detail diagram